PHENOLOGY AND NEST-SITE FIDELITY IN THREE SWIFT TOWERS IN THE LOMBARDY REGION

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The data used to write this poster was collected within the **"Home Swift Home" project**, monitoring three **artificial swift towers** in the provinces of Varese and Milan (Italy). The project started in 2018 monitoring the tower in **Jerago con Orago**, followed by **Azzate** and **Cuggiono**. Each tower contains multiple nest chambers, variably used by **Common Swifts (*Apus apus*)**.

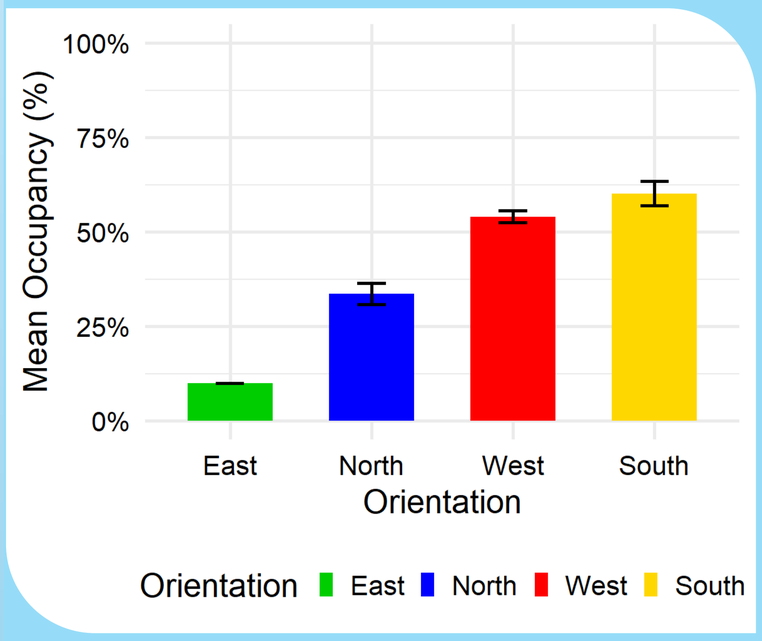
- The monitoring season started in middle to late April and, when possible, continued until fledging or tower vacancy.
- During monitoring each chamber was opened and the **number** of **eggs**, **chicks** and **adults** was recorded.
- Occasional events were also noted, such as mortality or occupation by other species.
- **Ringing sessions** were conducted to investigate nest- and mate-fidelity.
- Samples of feathers and *Crataerina pallida* (swift lousefly) found in the nests were also collected.



Logo by Giulia Belloni

OCCUPANCY AND EXPOSURE

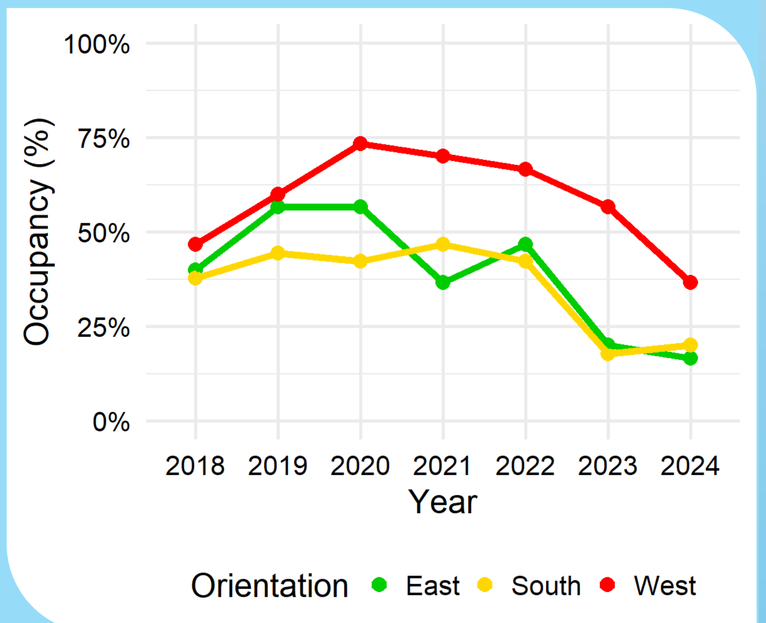
Analysis included Jerago con Orago and Cuggiono, since only one of the walls in the Azzate tower presents swift chambers.



In **Cuggiono**, **south-** and **west-facing chambers** were **consistently preferred** over north and east-facing ones throughout the two years of monitoring, indicating a **strong role of exposure in nest-site selection**.

In **Jerago con Orago**, **west-facing chambers** were significantly **more occupied**.

Occupancy declined due to rodent predation (observed during 2022).

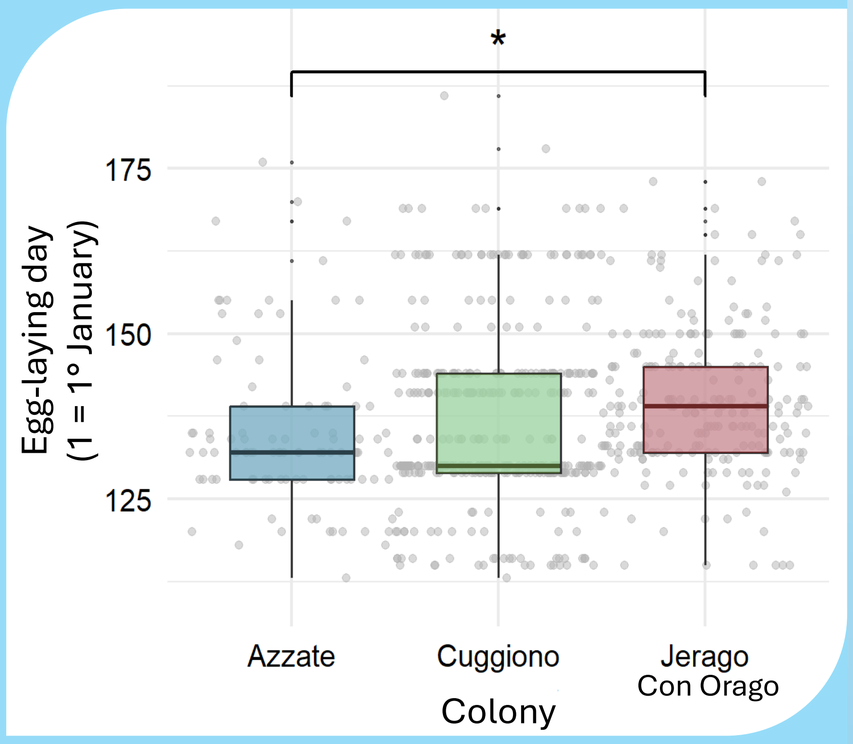


PHENOLOGY

Egg-laying dates were estimated as 20 days before the first chick observation (Majkusiak, 2022).

This approach was necessary since direct clutch observations often ended in failure.

We observed a **significant difference in egg-laying date** (ANOVA: X^2=6.76, df= 2, P=0.034) between Jerago con Orago and Azzate colonies.



NEST-SITE FIDELITY

Adult swifts show a strong fidelity to both the tower and the nest chamber.

- We used the percentage of **recaptures** over the total number of captures to measure **fidelity to the tower**;
- **Fidelity to the chamber** was calculated as the percentage of swifts captured in the same nest chamber as the **previous year** over the total number of recaptures.

The **sharp decline** of recaptures in the **Jerago con Orago** tower following 2022 is due to **rodent predation**.

Tower	Year	TOT. Capture	Capture	Recapture	Recaptures (% of TOT Capture)	Fidelity to chamber (% of Recaptures)
Azzate	*2022*	*19*	*19*	*/*	*/*	*/*
	2023	21	17	4	19,05%	75,00%
	2024	20	11	9	45,00%	14,29%
Cuggiono	*2022*	*41*	*41*	*/*	*/*	*/*
	2023	28	25	3	10,71%	66,67%
	2024	26	22	4	15,38%	100,00%
Jerago con Orago	*2018*	*49*	*49*	*/*	*/*	*/*
	2019	61	45	16	26,23%	68,75%
	2020	41	22	19	46,34%	73,68%
	2021	73	32	41	56,16%	63,41%
	2022	20	10	10	50,00%	62,50%
	2023	12	9	3	25,00%	0,00%
	2024	19	13	6	31,58%	0,00%

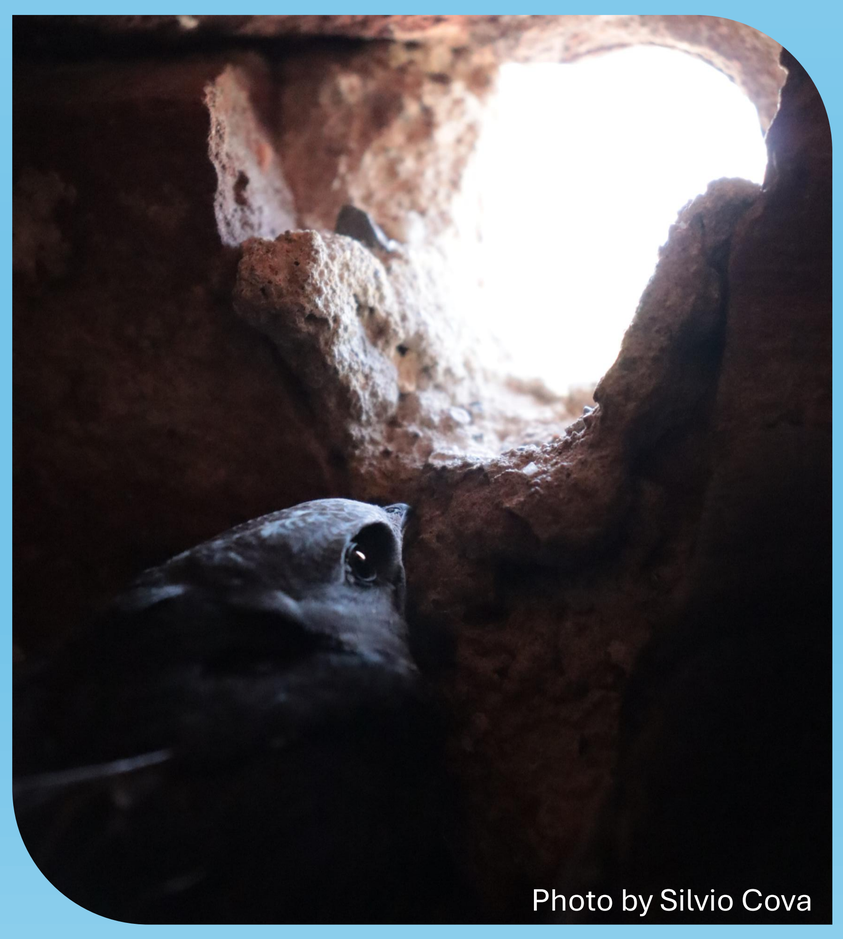


Photo by Silvio Cova

We would like to thank the Municipality of Jerago con Orago and the owners of the swift towers in Azzate and Cuggiono.
We are deeply grateful to the Insubric Group of Ornithology (GIO) and all its members, especially those that throughout the years made it so that this project could start and continue with their passion and effort.

References:

Boano G., Cucco M., Malacarne G., Orecchia G. (1993). Survival rate and mate fidelity in the Pallid Swift Apus pallidus. *Avocetta. 17:189-97.*

*Majkusiak, J. (2022). Breeding biology of the Common Swift (*Apus apus*) in Ireland – the most north-westerly edge of the nesting habitat.*

*Manica M., Casola D., Stocchetti A., Bazzarelli M., Biganzoli D., Morganti M. (2022). Breeding phenology, reproductive success and survival of Common Swift (*Apus apus*) from a Northern Italian colony. XIII EOU; Gießen (Germany).*

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